4 The Class Equation

Let G be a group and let

$$\mathcal{S}(G) = \{A : A \subset G, A \neq \emptyset\}$$

be the set of non-empty subsets of G. Then G acts on $\mathcal{S}(G)$ by conjugation via

$$\phi : G \times \mathcal{S}(G) \ \to \ \mathcal{S}(G)$$
$$(g, A) \ \mapsto \ gAg^{-1} = \{gag^{-1} \ : \ a \in A\}$$

where $g \in G$ and $A \subset G$ is a nonempty subset of G (check that this definition satisfies the axioms of a group action).

The subset gAg^{-1} of G is called the *conjugate* of A by g. Note that if A is a subgroup of G then gAg^{-1} is also a subgroup of A (check this fact).

Definition 4.1 (Normaliser)**.**
If $A \in \mathcal{S}(G)$, the stabiliser of A under the above action ϕ by conjugation is called the *Normaliser* of A in G and denoted

$$N_G(A) = \{g \in G \ : \ gAg^{-1} = A\}.$$

Thus $N_G(A)$ is a subgroup of G.

The normaliser $N_G(A)$ of a subset A of G acts on A by conjugation

$$\phi_A : N_G(A) \times A \to A$$
$$(g, a) \ \mapsto \ gag^{-1}$$

(check that ϕ_A satisfies the axioms of a group action).

Definition 4.2 (Centraliser-Centre)**.**
If $A \in \mathcal{S}(G)$, the kernel of the action ϕ_A is called the *centraliser* of A in G and denoted

$$C_G(A) = \{g \in G \ : \ gag^{-1} = a \quad \forall a \in A\}.$$

Thus $C_G(A)$ is a normal subgroup of $N_G(A)$ (see Definition 3.2). Note that the equality $gag^{-1} = a$ is equivalent to $ga = ag$, thus the centraliser of A in G

$$C_G(A) = \{g \in G \ : \ ga = ag \quad \forall a \in A\}$$

consists of those elements of G which commute with all elements of A.

In case $A = G$, $N_G(G) = G$ since $gGg^{-1} = G$ for all $g \in G$ (check this fact) and G is equal to its normaliser.

The centraliser $C_G(G)$ of G in G is called the *centre* of G and denoted

$$Z(G) = \{g \in G \ : \ gh = hg \quad \forall h \in G\}.$$

Thus $Z(G)$ is the set of elements of G which commute with every element of G. $Z(G)$ is a normal subgroup of G. Note that $Z(G)$ is contained in $C_G(A)$ for any non-empty subset A of G. Indeed an element of G which commutes with all elements of G commutes, a fortiori, with every element in a subset A of G. In fact

$$Z(G) = \bigcap_{A \subset G} C_G(A).$$

Note also that $Z(G)$ is an abelian group: elements of $Z(G)$ commute with each other.

Example 4.1.
(1) If G is abelian, all elements of G commute with each other: $ab = ba$ for all $a, b \in G$, so $Z(G) = G$. Similarly, $C_G(A) = N_G(A) = G$ for any subset A of G since $gag^{-1} = gg^{-1}a = a$ for every $g \in G$ and $a \in A$.

(2) Let $G = D_8$ be the dihedral group of cardinality 8 with the usual generators r and s satisfying $sr^j = r^{-j}s = r^{4-j}s$, and $A = \{1, r, r^2, r^3\}$ the subgroup of rotations in D_8 generated by r.

- We show $C_G(A) = A$. Since A is abelian: $r^i r^j = r^j r^i = r^{i+j}$, it holds that $A \subset C_G(A)$. The element $s \notin C_G(A)$ since $sr = r^3 s \neq rs$. An element of G not in A is of the form sr^i, and is not an element of $C_G(A)$ since $sr^i r = sr^{i+1} = r^{-1-i}s = r^{4-1-i}s \neq r^{i+1}s$ for $i = 0$ or $i = 2$. For $i = 1$ we have $(sr)r = sr^2 = r^2 s \neq r(sr) = rr^3 s = r^4 s = s$. Thus the only elements of $C_G(A)$ are the elements of A and $A = C_G(A)$.
- We show $N_G(A) = G$. Since $C_G(A) \subset N_G(A)$ [$C_G(A)$ is a subgroup of $N_G(A)$] and $A = C_G(A)$ we have $A \subset N_G(A)$. We have $sAs^{-1} = sAs = \{s1s = 1, srs = ssr^3 = r^3, sr^2 s = ssr^2 = r^2, sr^3 s = ssr = r\} = A$, hence $s \in N_G(A)$. Since $N_G(A)$ is a subgroup of G, $r^i \in N_G(A)$, $s \in N_G(A)$, it holds that $sr^i \in N_G(A)$, hence all elements of $G = D_8$ are in $N_G(A)$ and $N_G(A) = G$.
- We show $Z(G) = \{1, r^2\}$. Since $Z(G) \subset C_G(H)$ for any subset H of G, we have $Z(G) \subset C_G(A) = A$. The elements r, r^3 are not in $Z(G)$ since $rs = sr^3 \neq sr$ and $r^3 s = sr \neq sr^3$. Thus $Z(G) \subset \{1, r^2\}$. The element $r^2 \in Z(G)$ since it commutes with any power r^i of r, it also commutes with s: $sr^2 = r^2 s$, hence r^2 commutes with any element $s^i r^j$ of G. This shows $Z(G) = \{1, r^2\}$.

(3) Let $G = S_3$ and $A = \{1, (12)\}$. We show $C_G(A) = N_G(A) = A$. We have $A \subset C_G(A)$ since A is abelian. In particular A is a subgroup of $C_G(A)$ and 2 divides $|C_G(A)|$. By Lagrange's Theorem $|C_G(A)|$ divides $|G| = 6$, hence $|C_G(A)|$ equals 2 or 6. If $|C_G(A)| = 6$ then $G = C_G(A)$ and (12) would commute with every element of S_3. But this is not the case since (12) doesn't commute with (123): $(12)(123) = (23) \neq (123)(12) = (13)$. Thus, $|C_G(A)| = 2$ and $A = C_G(A)$.

Let $\sigma \in N_G(A)$. Then $\sigma A \sigma = \{\sigma 1 \sigma^{-1} = 1, \sigma(12)\sigma^{-1}\} = A$ holds if and only if $\sigma(12)\sigma^{-1} = (12)$, which implies $\sigma(12) = (12)\sigma$, meaning $\sigma \in C_G(A) = A$. Thus, $N_G(A) \subset C_G(A)$, hence $N_G(A) = C_G(A) = A$ since $C_G(A) \subset N_G(A)$ holds always. Finally $Z(G) = \{1\}$. Indeed $Z(G) \subset C_G(A) = A$. But $(12) \notin Z(G)$ since $(12)(123) = (23) \neq (123)(12) = (13)$. Thus $Z(G) = \{1\}$. We will see later that $Z(S_n) = \{1\}$ holds if $n \geq 3$.

4.1 The Class Equation

Let G be a group and $g \in G$. Consider the subset $\{g\} \subset G$, then

$$N_G(\{g\}) = C_G(\{g\}) = \{h \in G \ : \ hgh^{-1} = g\} = \{h \in G \ : \ hg = gh\}$$

is the subgroup of G of elements which commute with g. We denote $C_G(\{g\})$ simply by $C_G(g)$ and call it the *centraliser of g in G*. Note that $C_G(g) = \mathrm{Stab}(g)$ is precisely the stabiliser of $\{g\}$ under the conjugation action of G on itself.

The orbit of $\{g\}$ under conjugation is

$$\mathrm{orb}(g) = \{hgh^{-1} \ : \ h \in G\}$$

and consists of all elements of G which are conjugate to g; $\operatorname{orb}(g)$ is called the *conjugacy class* of g. Note that $g = e_G g e_G^{-1} \in \operatorname{orb}(g)$. Further $\operatorname{orb}(g) = \{g\}$ if and only if $hgh^{-1} = g$ [which is equivalent to $hg = gh$] for all $h \in G$, and this is equivalent to $g \in Z(G)$. Thus

$$|\operatorname{orb}(g)| = 1 \iff g \in Z(G).$$

Assume now that G is finite. The Orbit-Stabiliser Theorem (see Theorem 3.3) states that

$$|\operatorname{orb}(g)| = \frac{|G|}{|C_G(g)|} = |G : C_G(g)|.$$

The conjugacy classes of elements of G form a partition of G

$$G = \bigcup_{g \in G} \operatorname{orb}(g)$$

where the union is disjoint. By the above discussion, we have

$$Z(G) = \bigcup_{g \in G, |\operatorname{orb}(g)|=1} \operatorname{orb}(g)$$

where the union is over all element of G with $|\operatorname{orb}(g)| = 1$. Let $\{\operatorname{orb}(g_1), \ldots, \operatorname{orb}(g_r)\}$ be the distinct conjugacy classes of G which **are not** contained in $Z(G)$, or equivalently those orbits such that $|\operatorname{orb}(g_i)| \geq 2$ for $1 \leq i \leq r$. Then

$$G = Z(G) \bigcup \Big(\bigcup_{i=1}^{r} \operatorname{orb}(g_i) \Big).$$

Counting the number of elements of G, and bearing in mind the equality $|\operatorname{orb}(g_i)| = |G : C_G(g_i)|$, we find the class equation:

Theorem 4.1 (The Class Equation).
*Let G be a finite group and $\{\operatorname{orb}(g_1), \ldots, \operatorname{orb}(g_r)\}$ the distinct conjugacy classes of G which are **NOT** contained in $Z(G)$, then*

$$|G| = |Z(G)| + \sum_{i=1}^{r} |\operatorname{orb}(g_i)| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|.$$

Example 4.2.
(1) If G is abelian then all the elements of G commute, so $Z(G) = G$ and $|G| = |Z(G)|$. The class equation gives no specific information in this case.

(2) In D_8 we have $Z(D_8) = \{1, r^2\}$ [see example 4.1 (2)]. Let $x \in D_8$ and $x \notin Z(D_8)$. We show $|C_{D_8}(x)| = 4$. It holds that $\{1, r^2, x\} \subset C_{D_8}(x)$, since r^2 commutes with x and $C_{D_8}(x)$. Now $|C_{D_8}(x)|$ is a factor of $|D_8| = 8$, by Lagrange's Theorem. Given that $|C_{D_8}| \geq 3$, the only possibilities are 4 and 8. Further, $|C_{D_8}(x)| = 8$ cannot hold for otherwise $C_{D_8}(x) = D_8$ and x commutes with every element of D_8 hence $x \in Z(G)$ which is not the case. Thus $|C_{D_8}(x)| = 4$ necessarily, $|\operatorname{orb}(x)| = \frac{|D_8|}{|C_{D_8}(x)|} = |D_8 : C_{D_8}(x)| = 2$, and x has exactly two distinct conjugates.

The conjugacy classes of D_8 are

$$\{1\},\ \{r^2\},\ \{r, r^3\},\ \{s, sr^2\},\ \{sr, sr^3\}.$$

To see this, note that $r^3 = srs$, $sr^2 = (sr)s(r^3s)$, and $sr^3 = (sr^2)sr(r^2s)$.

The first two classes form $Z(D_8)$ and the class equation for D_8 is

$$|D_8| = 8 = 1 + 1 + 2 + 2 + 2.$$

One important consequence of the Class equation is the following.

Theorem 4.2.
If p is a prime number and G is a finite group of cardinality $|G| = p^s$ for some $s \geq 1$, then G has a nontrivial centre meaning $|Z(G)| > 1$.

Proof. By the class equation

$$|G| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|$$

where $g_1, \ldots, g_r$ are representative of the distinct conjugacy classes not contained in $Z(G)$. By definition $C_G(g_i) \neq G$ since $g_i \notin Z(G)$, hence $|G : C_G(g_i)| = \frac{|G|}{|C_G(g_i)|} > 1$, and p divides $|G : C_G(g_i)|$ (since the latter is > 1 and divides $|G| = p^s$). Since p divides $|G|$ it must also divide

$$|Z(G)| = |G| - \sum_{i=1}^{r} |G : C_G(g_i)|,$$

in particular $|Z(G)| > 1$ and $Z(G) \neq \{e_G\}$. □

4.2 Cauchy's Theorem

Using among others the class number formula we now prove Cauchy's Theorem.

Theorem 4.3 (Cauchy's Theorem)**.**
Let G be a finite group and p a prime number which divides $|G|$. Then there exists an element of G of order p, and a subgroup of G of cardinality p.

Proof. The proof is by strong induction on $|G|/p$. If $|G| = p$, then G is cyclic of order p and every element $x \in G$ different from e_G has order p. So assume $|G| > p$, and p divides $|G|$.

First, we treat the case where G is **abelian**. Let $x \in G$ be an element distinct from e_G and $n = \text{ord}(x) > 1$. If p divides n, and $m = \frac{n}{p}$, then x^m is an element of order p. Otherwise, if $\gcd(p, n) = 1$, let $H = \langle x \rangle$ be the subgroup generated by x; which is normal since G is abelian. Consider the factor group G/H whose cardinality is smaller than that of G, and is divisible by p since p doesn't divide $|H| = n$ and divides $|G| = |H||G/H|$. By the induction hypothesis G/H has an element yH of order p. But yH is the image of y under the natural homomorphism $G \to G/H$. Thus, $p = \text{ord}(yH)$ divides $r = \text{ord}(y)$ and $y^{\frac{r}{p}}$ is an element of G of order p.

Next, we treat the general case where G need not be abelian. If p divides $|Z(G)|$ then, since $Z(G)$ is abelian, we deduce from the above that there exists an element x of $Z(G)$, hence an element of G, such that $\text{ord}(x) = p$. Otherwise consider the class equation

$$|G| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(g_i)|$$

where $\{\mathrm{orb}(g_1), \ldots, \mathrm{orb}(g_r)\}$ are the distinct conjugacy classes of G which are not contained in $Z(G)$. Now p divides $|G|$, and doesn't divide $|Z(G)|$, hence there exists $1 \leq i \leq r$ such that p doesn't divide $|G : C_G(g_i)|$ (otherwise p would divide $|G| - \sum_{i=1}^{r} |G : C_G(g_i)| = |Z(G)|$). This implies that p divides $|C_G(g_i)| = \frac{|G|}{|G:C_G(g_i)|}$. Now $C_G(g_i)$ is different from G, since g_i is not in $Z(G)$, and p divides $|C_G(g_i)| < |G|$. By the induction hypothesis there exists an element $x \in C_G(g_i)$; hence a fortiori an element of G, such that $\mathrm{ord}(g) = p$.

Finally, if $x \in G$ has order p then the subgroup $\langle x \rangle$ generated by x is a subgroup of G of cardinality p. □

4.3 Conjugacy Classes in S_n

We have already seen that two permutations $\sigma, \tau \in S_n$ are conjugate if and only if they have the same cycle type (see Proposition 2.5).

Let $n \geq m \geq 2$ be integers and $\sigma = (a_1 \ldots a_m)$ an m-cycle in S_n. There are n-possibilities for a_1. Once a_1 is fixed there are $n-1$ possibilities for a_2, and so on. Once $a_1, \ldots, a_{m-1}$ are fixed there are $n - (m-1) = n - m + 1$ choices for a_m. Shifting the position of $a_1, \ldots, a_m$ cyclically to obtain $(a_i a_{i+1} \ldots a_m a_1 a_2 \ldots a_{i-1})$ doesn't change the permutation σ. Counting in this way the m-cycles we find that their number is

$$\frac{n(n-1)\ldots(n-m+1)}{m}.$$

The number $\frac{n(n-1)\ldots(n-m+1)}{m}$ is thus the cardinality of the orbit $\mathrm{orb}(\sigma)$ of σ under conjugation by S_n, which is also the index of the centraliser of σ in S_n:

$$|\,\mathrm{orb}_{S_n}(\sigma)| = \frac{|S_n|}{|C_{S_n}(\sigma)|} = |S_n : C_{S_n}(\sigma)| = \frac{n(n-1)\ldots(n-m+1)}{m}.$$

Since $|S_n| = n!$ we deduce that

$$|C_{S_n}(\sigma)| = \frac{|S_n|}{\frac{n(n-1)\ldots(n-m+1)}{m}} = \frac{n!}{\frac{n(n-1)\ldots(n-m+1)}{m}} = m(n-m)!.$$

The elements $\{1, \sigma, \ldots, \sigma^{m-1}\}$, which form the subgroup of S_n generated by σ, all commute with σ. Also all permutations in S_n which are disjoint from σ commute with σ and there are $(n-m)!$ permutations of this type: the full symmetric group on the digits in $\{1, 2, \ldots, n\}$ not appearing in σ, i.e. distinct from $\{a_1 \ldots a_m\}$.

The product of elements of these two types already account for $m(n-m)!$ permutations commuting with σ. By the above computation of $|C_{S_n}(\sigma)|$ these form the full centraliser of σ in S_n. Summarising, if $\sigma \in S_n$ is an m-cycle, then

$$C_{S_n}(\sigma) = \left\{\sigma^i \tau \ : \ 0 \leq i \leq m-1, \ \tau \in S_{n-m}\right\}$$

where S_{n-m} denotes the subgroup of S_n which fixes all integers appearing in the m-cycle σ [this subgroup is isomorphic to the symmetric group on $n-m$ letters]: these are precisely the permutations of S_n which are disjoint from σ.

For example the centraliser of $\sigma = (135)$ in S_7 is the subgroup

$$\left\{(135)^i \tau \ : \ 0 \leq i \leq 2 \quad \text{and } \tau \text{ fixes } 1, 3, 5\right\}.$$

Note that $\tau \in S_A$ where $A = \{2, 4, 6, 7\}$, so there are $4!$ choices for τ and S_A is isomorphic to S_4. In this case the centraliser $C_{S_7}(\sigma)$ has cardinality $3.4! = 72$.

4.4 Simple Groups

A finite group G is called **simple** if $|G| > 1$ and the only normal subgroups of G are $\{e_G\}$ and G itself.

By Lagrange's Theorem, if $|G| = p$ is a prime number, then the only subgroups of G are G and $\{e_G\}$, so G is necessarily simple in this case. Thus all cyclic groups with cardinality a prime number are simple. In fact every abelian simple group G is isomorphic to $\mathbb{Z}/p\mathbb{Z}$ for some prime number p (think how to prove this fact as an exercise).

An important example of a finite simple non abelian group is the following.

Theorem 4.4.
The alternating group A_5 is simple.

Proof. First, note that a normal subgroup H of a group G is the union of the conjugacy classes of its elements. Indeed, whenever H contains an element g of G it contains its orbit $\mathrm{orb}(g)$ under conjugation. Hence if H is normal it holds that

$$H = \bigcup_{g \in H} \mathrm{orb}(g)$$

where the union is disjoint.

We work out the conjugacy classes of elements of A_5 and their cardinalities. Remember that permutations which are conjugate in S_n need not be conjugate in A_n [see example 2.3 (4)] so we need to be careful.

Representatives of the conjugacy classes of even permutations in S_5 can be taken to be

$$1 \qquad (123) \qquad (12345) \qquad \text{and} \qquad (12)(34).$$

The centralisers of 3-cycles and 5-cycles in S_5 have been determined above. We have

$$C_{S_5}((123)) = \{(123)^i \tau \ : \ 0 \leq i \leq 2 \quad \text{and } \tau \text{ fixes } 1, 2, 3\},$$

thus τ is either identity or the transposition (45), and

$$C_{S_5}((12345)) = \{(12345)^i \ : \ 0 \leq i \leq 4\}.$$

Checking which of these conjugates are even we find

$$C_{A_5}((123)) = C_{S_5}((123)) \cap A_5 = \langle (123) \rangle = \{(123)^i \ : \ 0 \leq i \leq 2\}$$

and

$$C_{A_5}((12345)) = C_{S_5}((12345)) \cap A_5 = \{(12345)^i \ : \ 0 \leq i \leq 4\}.$$

These groups have cardinalities 3 and 5, hence have index 20 and 12 in A_5, respectively. Thus there are 20 distinct conjugates of (123) in A_5 and 12 distinct conjugates of (12345) in A_5:

$$|\,\mathrm{orb}_{A_n}((123))| = 20$$

$$|\,\mathrm{orb}_{A_n}((12345))| = 12.$$

Since there are twenty 3-cycles in S_5, and all of them are in A_5, we see that all twenty 3-cycles are conjugate in A_5.

The number of 5-cycles in A_5 is 24, but there are only 12 distinct conjugates of the 5-cycle (12345) in A_5. Thus some 5-cycle σ is not conjugate to (12345) in A_5 [in fact (13245) is such a σ]. Hence the 5-cycles in A_5 form two distinct conjugacy classes in A_5, each of these classes has 12 elements.

Since the 3 and 5-cycles account for all the nonidentity elements of A_5 with odd order, the 15 remaining nonidentity elements of A_5 must have even order and therefore have cycle type $(2, 2)$.

Certainly, the conjugacy class of $(12)(34)$ in S_5 consists of all permutations of cycle type $(2, 2)$. We count these. Let σ be a permutation of cycle type $(2, 2)$. There are exactly five choices for i such that $\sigma(i) = i$. Suppose σ acts non-identically on $\{j_1, \ldots, j_4\}$. Then there are 3 choices for $\sigma(j_1)$. This fixes σ. So there are $5 \cdot 3 = 15$ choices for σ.

Now I'll show that every $(2, 2)$ cycle is conjugate to $\tau = (12)(34)$ in A_5. Suppose σ is a permutation of cycle type $(2, 2)$. Then, $\sigma = g\tau g^{-1}$, for some $g \in S_5$. If g is even, then σ is conjugate to τ in A_5. Otherwise, noting that (12) commutes with τ, we have

$$(g(12))\tau(g(12))^{-1} = g\tau g^{-1} = \sigma.$$

But $g(12)$ is even, so σ is conjugate to τ in A_5. Thus, $|\operatorname{orb}_{A_n}((12)(34))| = 15$.

In summary the conjugacy classes of A_5 have orders 1, 15, 20, 12 and 12.

Now suppose H is a normal subgroup of A_5, it must be the union of conjugacy classes of A_5. Thus the cardinality of H must be both a divisor of 60 (the cardinality of A_5) and the sum of some collection of the integers $\{1, 12, 12, 15, 20\}$ (the sizes of the conjugacy classes in A_5). A quick check shows that the only possibilities are $|H| = 1$ and $|H| = 60$, so that H is either equal to $\{1\}$ or A_5 itself. □

A more general result holds, namely the alternating group A_n is simple for every $n \geq 5$. We omit the proof of this fact.